# Clean Control Group – Contractor Experience

2024-08-22

- Let $n_i^{task}$ = Number of projects of the same task held by the contractor in the three quarters before starting project $i$
- Let $n_i$ = Number of projects held by the contractor in the three quarters before starting project $i$
- Define:
  - Inexperienced contractor$_i = 1$ if $n_i^{task} = 0$
  - New entrant$_i = 1$ if $n_i = 0$
  - Level of inexperience$_i = 1 - \frac{n_i^{task}}{n_i}$ if $n_i > 0$
  - Level of inexperience$_i = 1$ if $n_{i,k,t} = 0$.

## 1 Regression results

- For $IE + Treat \times IE + IE \times Post + Treat \times IE \times Post$ using the two discrete metrics, we get:

| | Estimate | SE | 2.5 % | 97.5 % | Hypothesis | z value | Pr(>\|z\|) |
|---|---|---|---|---|---|---|---|
| Task inexperienced | 1.15 | 0.13 | 0.89 | 1.42 | 0.00 | 8.60 | 0.00 |

| | Estimate | SE | 2.5 % | 97.5 % | Hypothesis | z value | Pr(>\|z\|) |
|---|---|---|---|---|---|---|---|
| New entrant | 1.08 | 0.18 | 0.73 | 1.44 | 0.00 | 5.98 | 0.00 |

Table 1: Logit model

| | IE | | |
|---|---|---|---|
| | Task inexperienced | New entrant | Level of inexperience |
| | I | II | III |
| | Logit | Logit | OLS |
| Treat | 0.201*** | 0.271*** | -0.014*** |
| | (0.037) | (0.047) | (0.004) |
| Treat × Post | 0.055 | 0.200*** | 0.028*** |
| | (0.036) | (0.043) | (0.004) |
| Time FE | Yes | Yes | Yes |
| Task FE | Yes | Yes | Yes |
| Industry FE | Yes | Yes | Yes |
| Observations | 156,327 | 162,648 | 165,567 |
| Squared Correlation | 0.30 | 0.21 | 0.30 |
| Pseudo $R^2$ | 0.25 | 0.21 | 0.43 |
| BIC | 157,141.70 | 118,828.23 | 95,229.34 |

*Clustered (Project ID) standard-errors in parentheses*
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

| | Percentage delay rate | | |
|---|---|---|---|
| | Task inexperienced | New entrant | Level of inexperience |
| | I | II | III |
| Treat | -0.497*** | -0.633*** | -0.297 |
| | (0.155) | (0.130) | (0.284) |
| IE | 1.414*** | 1.310*** | 1.264*** |
| | (0.184) | (0.228) | (0.282) |
| Treat × IE | -0.585*** | -0.736*** | -0.550* |
| | (0.206) | (0.270) | (0.323) |
| IE × Post | -0.095 | 0.043 | 0.327 |
| | (0.257) | (0.339) | (0.352) |
| Treat × Post | 0.806*** | 0.917*** | 1.057*** |
| | (0.188) | (0.163) | (0.337) |
| Treat × IE × Post | 0.418 | 0.465 | -0.211 |
| | (0.301) | (0.409) | (0.413) |
| Time FE | Yes | Yes | Yes |
| Task FE | Yes | Yes | Yes |
| Industry FE | Yes | Yes | Yes |
| Observations | 165,473 | 165,473 | 165,473 |
| $R^2$ | 0.22 | 0.22 | 0.22 |
| Within $R^2$ | 0.15 | 0.15 | 0.15 |

*Clustered (Project ID) standard-errors in parentheses*
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*